AYOUB KHALAL

Actuarial Science & Data Science | Artificial Intelligence | Digital Infrastructure

Ifrane, Morocco | +212 614 230 328 | ayoub.khalal01@gmail.com **| LinkedIn:** Ayoub Khalal **| GitHub:** ayoubkhalal01-lab **| Portfolio:** ayoubkhalal.tech

PROFESSIONAL SUMMARY

Master's student in Actuarial Science and Data Science (Excellence Program), specialized in data analysis, statistics and artificial intelligence applied to actuarial science. Strong expertise in insurance pricing and risk management (life, non-life, reinsurance, pension planning, group insurance), combined with data science skills: data extraction, cleaning, transformation and visualization, machine learning and predictive modeling with Python (Pandas, NumPy, Scikit-learn) and R. Analytical rigor, research mindset, and solid foundations in descriptive and inferential statistics, risk theory and credibility theory. Actively seeking an internship opportunity in Data Science, Artificial Intelligence, Machine Learning or Actuarial Science.

TECHNICAL SKILLS

Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, TensorFlow, Machine Learning, Deep Learning, Big Data, data extraction, cleaning, transformation and visualization, claims prediction, customer scoring, segmentation

Statistics & Actuarial Science: Descriptive and inferential statistics, Bayesian methods, time series analysis, advanced probability, statistical inference methods, risk theory, decision theory, credibility theory, survival analysis, insurance pricing, liability modeling, technical reserves calculation

Algorithmic Structures: Lists, trees, graphs

Systems & Networks: Linux, Windows Server, Shell/PowerShell scripting, Cisco configuration (CCNA), LAN maintenance and security

Cybersecurity: Threat analysis, firewalls, security policies

Cloud Computing: Azure, Google Cloud

Regulation & Finance: Regulatory and financial understanding of insurance markets, financial and insurance risk management, financial econometrics and forecasting, financial market analysis, insurance and banking accounting, Financial Technology, Blockchain

TOOLS & TECHNOLOGIES

Programming: Python, SQL, R, Java, Bash

Data Science: Jupyter, JupyterLab, Google Colab, Kaggle, Anaconda, Pandas, NumPy, SciPy, Scikit-learn, Matplotlib, Seaborn, Plotly, Excel, Tableau

Artificial Intelligence: TensorFlow, PyTorch, Hugging Face, Ollama, Prompt Engineering, RAG (Retrieval-Augmented Generation), Embeddings, Vector Databases, Model fine-tuning

Big Data: Apache Spark, PySpark, Apache Hadoop, HDFS, MapReduce

Actuarial Science: Python, SPSS, Excel, Power BI, SQL

Databases: MySQL

Software Development: VS Code, Git, GitHub, Docker, Windows Terminal, Ubuntu (WSL)

Networks & Infrastructure: Cisco Packet Tracer, GNS3, Wireshark, PuTTY, VMware, VirtualBox, OpenStack

Cloud: Azure, Azure ML, Google Cloud Platform

Deployment: FastAPI, Flask, Streamlit

ACADEMIC & PERSONAL PROJECTS

- NVIDIA Stock Price Forecasting – Time series forecasting model
- Bank Customer Risk Classification – Machine Learning model predicting customer risk level
- HR & Labor Law AI Assistant – Conversational AI agent for HR and labor law questions

EDUCATION

Master's Degree (Excellence Program) in Actuarial Science and Data Science *November 2025*
FEG, Settat

Bachelor's Degree (Excellence Program) in Actuarial Science and Artificial Intelligence *November 2024 – June 2025*
FEG, Settat

Digital Infrastructure Diploma — Networks and Systems Track *September 2022 – June 2024*
OFPPT, Ifrane

PROFESSIONAL EXPERIENCE

Intern — GAO Tek (New York, USA – Remote) *August 2026 – November 2026 (ongoing)*

- Contributing to data collection and data mining for company projects

Intern — CNSS (National Social Security Fund) *July 2025 – August 2025*

- Managed insured members' files under the social security scheme
- Performed control and verification of data related to insured members
- Contributed to digitalization initiatives for administrative processes

Intern — Ifrane Municipality *March 2024 – April 2024*

- Designed an HR database for human resources management
- Managed and maintained network equipment
- Contributed to information system security

CERTIFICATIONS

- Certiport Word Office
- Certificate of Excellence — ISTA Ifrane

LANGUAGES

Arabic, French, English